CHAPTER 24 · DERIVATIONS

QUANTUM PHYSICS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 RADII OF BOHR ORBITS

SYLLABUS 25.3(B)

RESULT $r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}$

1. The electric attraction provides the centripetal force: $\dfrac{e^2}{4\pi\varepsilon_0 r^2} = \dfrac{mv^2}{r}$ … (1)
2. Bohr's postulate (quantised angular momentum): $mvr = \dfrac{nh}{2\pi}$, so $v = \dfrac{nh}{2\pi mr}$ … (2)
3. Substitute (2) into (1): $\dfrac{e^2}{4\pi\varepsilon_0 r} = mv^2 = \dfrac{n^2h^2}{4\pi^2 mr^2}$.
4. $\mathbf{r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}}$; $r_1 = 5.3 \times 10^{-11}$ m.

2 ENERGY LEVELS OF HYDROGEN

SYLLABUS 25.3(C)

RESULT $E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}$

1. From (1): $KE = \tfrac{1}{2}mv^2 = \dfrac{e^2}{8\pi\varepsilon_0 r}$.
2. $PE = -\dfrac{e^2}{4\pi\varepsilon_0 r}$.
3. Total $E = KE + PE = -\dfrac{e^2}{8\pi\varepsilon_0 r}$.
4. Substitute r_n: $\mathbf{E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}} = -\dfrac{13.6}{n^2}$ eV.

3 MINIMUM X-RAY WAVELENGTH

SYLLABUS 25.4(C)

RESULT $\lambda_{min} = \dfrac{hc}{eV}$

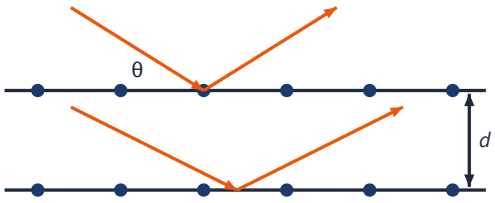
1. Electrons accelerated through p.d. V gain kinetic energy eV.
2. The shortest wavelength is emitted when all of this energy goes into one photon: $eV = hf_{max} = \dfrac{hc}{\lambda_{min}}$.
3. $\mathbf{\lambda_{min} = \dfrac{hc}{eV}}$ (independent of the target metal).

4 BRAGG'S LAW

SYLLABUS 25.4(E)

RESULT $2d \sin \theta = m\lambda$

1. X-rays are reflected by two adjacent atomic planes a distance d apart, at glancing angle θ.
2. The ray from the lower plane travels an extra path 2d sin θ.
3. Constructive interference when the extra path is a whole number of wavelengths: **$2d \sin \theta = m\lambda$**.



extra path of lower ray = $2d \sin \theta$